

Rotor Aerodynamics

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46110 Fundamentals of Aerodynamics

In this chapter the basics of axial momentum theory is described and applied to describe the aerodynamics of rotors. The chapter is divided into subchapters describing the theory of helicopters, aircraft propellers and wind turbine rotors, respectively. As will be shown in the following, the three types of rotary machinery contain many similarities, but they also have different symbol conventions and peculiarities with respect to their design and operating envelopes that makes it convenient to divide their descriptions into separate subsections. Most of the theory outlined in the following is taken from the textbooks of Glauert (1935), Leishman (2000) and Sørensen (2016).

The basic tool for understanding rotary aerodynamics is the momentum theory in which the flow is assumed to be steady, inviscid, incompressible and axisymmetric. The momentum theory basically consists of control volume integrals for conservation of mass, axial and angular momentum balances, and energy conservation,

$$\oint_{CV} \rho \mathbf{V} \cdot d\mathbf{A} = 0 \quad (1)$$

$$\oint_{CV} u \rho \mathbf{V} \cdot d\mathbf{A} = T - \oint_{CV} p d\mathbf{A} \cdot \mathbf{e}_x \quad (2)$$

$$\oint_{CV} r u_\theta \rho \mathbf{V} \cdot d\mathbf{A} = Q \quad (3)$$

$$\oint_{CV} \left[p / \rho + \frac{1}{2} \|\mathbf{V}\|^2 \right] \rho \mathbf{V} \cdot d\mathbf{A} = P \quad (4)$$

where $\mathbf{V} = (u, v, w)$ is the velocity vector in axial, radial and azimuthal direction, respectively, ρ is the density of air, $\mathbf{A}$ denotes the outward pointing area vector of the control volume, p is the pressure, T is the axial force (thrust) acting on the rotor, Q is the torque and P is the power required to operate the rotor. These equations form the basics of most of the theory presented in the following.

1. Helicopter aerodynamics

The helicopter constitutes one of the most complex aircraft machineries in aerodynamics. Its operational envelope includes both hover, climb/descent and forward flight, as well as various complex maneuvers in three dimensions. In brief, a helicopter rotor constitutes three main functions, which are to generate a thrust to outbalance the weight of the helicopter, a propulsive force to ensure a forward motion and means to generate forces and moments for various maneuvers in three dimensions. Because of the motion of the propeller and the different flight conditions, aerodynamics is associated with complicated dynamic features as well as compressibility effects. However, the essential part of the flight duties, such as hover, climb, and steady forward flight, can be explained and analyzed by using steady flow

approximations. In the following, we go through the main characteristics of the helicopter aerodynamics, including hover, climb/descent, and forward flight, and devise some simple relations for the design of the rotor as well as analysis tools based on blade-element momentum theory.

1.1 Dimensionless parameters

The main dimensionless parameters to characterize the aerodynamic operation of a helicopter are the following,

$$\text{Thrust coefficient: } C_T = \frac{T}{\frac{1}{2}\rho A(\Omega R)^2} \quad (5)$$

$$\text{Power coefficient: } C_P = \frac{P}{\frac{1}{2}\rho A(\Omega R)^3} \quad (6)$$

where Ω is the angular velocity of the rotor, A is the rotor area, and R is the radius of the rotor. Note that in the US it is customary to define the coefficients without the one half factor in the denominator, hence they take half the value as those defined in the present work, which is the traditional definition in Europe.

1.2 The aerodynamics of a hovering rotor

A helicopter spends most of its time in hover, which is the flight condition for which the essential part of the rotor is designed. In hover, the main purpose of the rotor is to produce a vertical force, the thrust, in order to outbalance the weight of the helicopter. In the following, axial momentum theory will be used to make a first basic analysis of the power to keep the helicopter in the air. Next, losses will be included and some simple equations derive to make a first overall layout of the rotor. Finally, some simple design rules will be given for designing the plan form and twist distribution of the rotor.

1.2.1 Axial momentum theory

In the following, axial momentum theory will be employed to give an overall estimate of the aerodynamics of a hovering helicopter. This is accomplished by considering the flow going through the rotor plane of the helicopter and defining a control volume consisting of the stream surface going through the tips of the rotor blades, as shown in Fig. 1. Since no flow (per definition) can transverse a stream surface, the mass flow is constant across the planes parallel to the rotor plane. In Fig. 1, the dashed cross section is an upstream plane, which is to be located so far away from the helicopter that the flow is unaffected by the presence of the hovering rotor. Hence, in this plane the incoming velocity, U_0 , is so small that it can be neglected, i.e. $U_0 \approx 0$. In order to keep the helicopter still in the air an upward pointing thrust force, T , is created, which will generate an induced velocity, v_i , in the rotor plane.

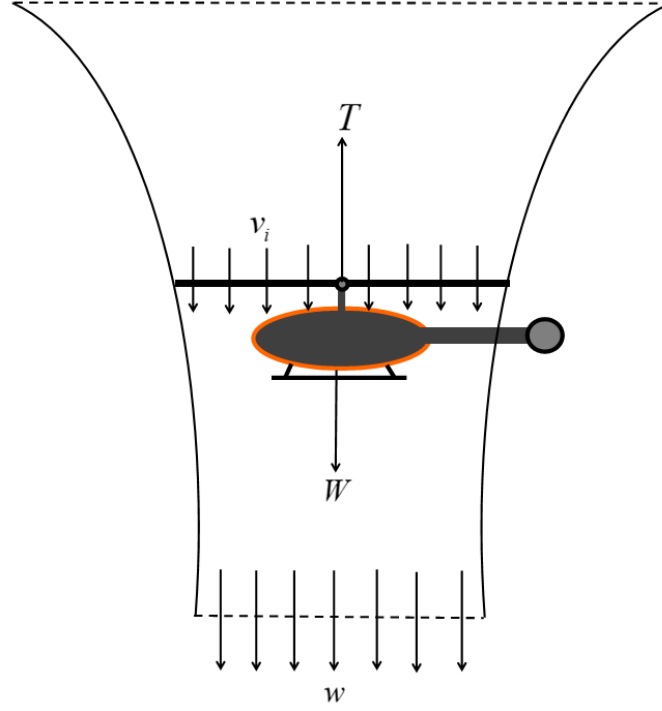


Fig. 1. Control volume and flow characteristics of actuator disc model of hovering rotor.

Denoting by $\dot{m}$ the mass flow through the rotor plane, mass conservation (eq. 1) gives the following integral relations,

$$\dot{m} = \int_{disc} \rho \mathbf{V} \cdot d\mathbf{A} = \int_{\infty} \rho \mathbf{V} \cdot d\mathbf{A}, \quad (7)$$

where subscript ‘disc’ refers to the rotor plane and ‘ ∞ ’ to the downstream outflow plane. Assuming one-dimensional flow theory and denoting the by w the outflow velocity in the wake, we get the following relationship,

$$\dot{m} = \rho A v_i = \rho A_{\infty} w, \quad (8)$$

where A_{∞} denotes the (unknown) area of the outflow plane.

A main approximation in momentum theory is to replace the action of the rotor on the flow with an actuator disc. An actuator disc is an idealized rotor representation, in which a sudden pressure difference is created over the rotor without having any discontinuity of the velocity. For a hovering helicopter, a low pressure, p^- , appears in front of the disc whereas a higher pressure, p^+ , acts behind the rotor disc. The pressure jump over the rotor disc, $(p^+ - p^-)$, then corresponds to the thrust force acting over the rotor, $T = (p^+ - p^-)A$, where A is the area of the rotor. The extracted power can then be determined by multiplying the thrust by the local velocity in the rotor plane. The thrust has to be in balance with the weight of the rotor in order to keep the helicopter still in the air, hence we get that $T = mg$, where m is the mass of the helicopter and g is the gravitational acceleration.

Applying axial momentum balance over the actuator disc, eq. (2), utilizing the control volume in Fig. 1, we get

$$T = \int_{\infty} u \rho \mathbf{V} \cdot d\mathbf{A} - \int_0 u \rho \mathbf{V} \cdot d\mathbf{A}, \quad (9)$$

where the contribution of the pressure disappears as it can be shown that its net contribution on the control volume is zero. Assuming zero negligible inflow velocity, i.e. $U_0 \approx 0$, the second integral becomes zero, and we get

$$T = \int_{\infty} u \rho \mathbf{V} \cdot d\mathbf{A} = \dot{m} w. \quad (10)$$

From energy conservation, eq. (4), the consumed power reads

$$P = T v_i = \int_{\infty} \frac{1}{2} u^2 \rho \mathbf{V} \cdot d\mathbf{A} - \int_0 \frac{1}{2} u^2 \rho \mathbf{V} \cdot d\mathbf{A}, \quad (11)$$

Resulting in the following relationship.

$$T v_i = \int_{\infty} \frac{1}{2} u^2 \rho \mathbf{V} \cdot d\mathbf{A} = \frac{1}{2} \dot{m} w^2. \quad (12)$$

Multiplying eq. (10) by v_i , and combining it with eq. (12), the following relationship is obtained,

$$\dot{m} v_i w = \frac{1}{2} \dot{m} w^2 \Rightarrow v_i = \frac{1}{2} w, \quad (13)$$

showing that the induced velocity in the rotor plane is equal to half the value of the induced velocity in the far wake. Exploiting this in eq. (10), gives the result,

$$T = \dot{m} w = 2 \rho A v_i^2. \quad (14)$$

Hence, expressed in terms of the thrust, the induced velocity reads

$$v_i = \sqrt{\frac{T}{2 \rho A}}, \quad (15)$$

from which we get

$$P = T v_i = \frac{T^{3/2}}{\sqrt{2 \rho A}}. \quad (16)$$

Utilizing that the weight of a helicopter in hover is in balance with the produced thrust, i.e. $T = W = mg$, the above equation may be written as

$$P = \frac{(mg)^{3/2}}{\sqrt{2\rho A}}, \quad (17)$$

where m is the mass of the helicopter and g is gravity constant. This expression shows that the required power to operate a hovering helicopter is inversely proportional to the area swept by the rotor. Hence, the rotor should be equipped with long and slender blades. It should be mentioned, however, that these expressions are derived without losses and, as will be shown later, adding friction and tip losses will tend to limit the length of blades.

1.2.2 Blade–element analysis and overall rotor layout

As shown in Leishman (2000), a simple approximate analysis makes it possible to include the main losses. Hence, friction losses may be incorporated by making a simplified blade–element analysis, enabling the development of a simple equation to make a first rotor layout including losses. Consider a cross section of radial extent, dr , of a rotor blade (see Fig. 2), the local axial and tangential force components per length unit may be written as, respectively,

$$f_z = \frac{dF_z}{dr} = f_L \cos \phi - f_D \sin \phi, \quad (18)$$

$$f_x = \frac{dF_x}{dr} = f_L \sin \phi + f_D \cos \phi, \quad (19)$$

where f_L is the local lift component, f_D is the local drag component, and ϕ is the flow angle.

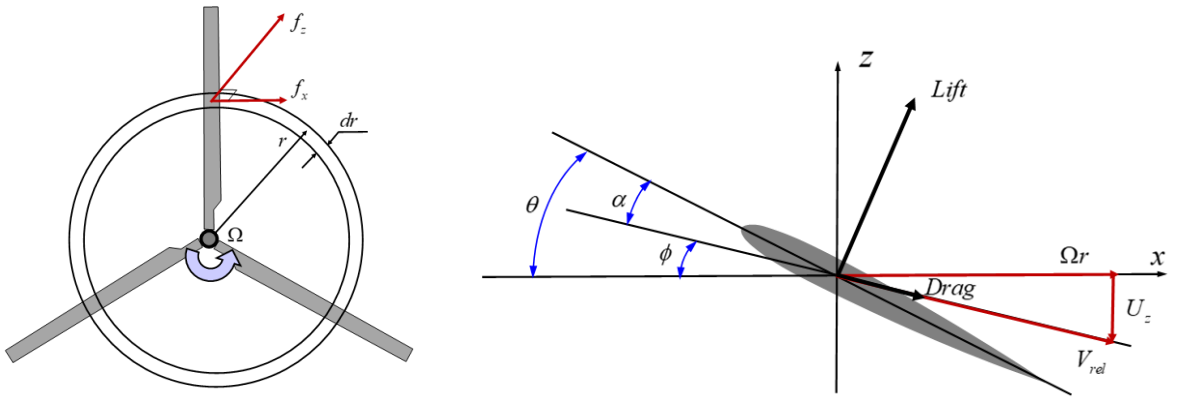


Fig. 2. Geometry and definition of forces and airfoil angles.
Left: Rotor system; Right: Airfoil coordinates.

Assuming the rotor to consist of N_b rotor blades, at a radial distance, r , from the rotor axis, the thrust and resulting torque components write

$$dT = N_b dF_z = N_b (f_L \cos \phi - f_D \sin \phi) dr \quad (20)$$

$$dM_z = N_b r dF_x = N_b r (f_L \sin \phi + f_D \cos \phi) dr. \quad (21)$$

Multiplying the torque by the rotational speed, Ω , gives the contribution of the blade element to the power consumption,

$$dP = \Omega dM_z = N_b \Omega r dF_x = N_b \Omega r (f_L \sin \phi + f_D \cos \phi) dr. \quad (22)$$

From eq. (20), it is seen that the drag contributes negatively to the thrust whereas it is added as an additional expense to the power consumption. Hence, drag, which essentially is due to the friction in the boundary layer, represents a loss of energy. Denoting the additional power to due to friction loss as P_0 , and assuming small flow angles, i.e. $\cos \phi \approx 1$, one obtains,

$$dP_0 = N_b \Omega r f_D \cos \phi dr \approx N_b \Omega f_D r dr. \quad (23)$$

Introducing a drag coefficient, C_{d0} , and assuming small induced velocities as compared to the local rotational velocity, the drag writes

$$f_D \approx \frac{1}{2} \rho c (\Omega r)^2 C_{d0}. \quad (24)$$

Inserting this into eq. (23), assuming constant chord, c , and drag coefficient, integration over the rotor blades gives the following simple expression for the friction loss,

$$P_0 \approx \frac{1}{2} \rho c N_b \Omega^3 C_{d0} \int_0^R r^3 dr = \frac{1}{8} \rho c N_b C_{d0} \Omega^3 R^4. \quad (25)$$

Additional losses due to tip losses, wake losses, etc, may be incorporated through the so-called induced power correction factor, γ , resulting in the following equation for the power consumption,

$$P = \gamma P_{ideal} + P_0, \quad (26)$$

where

$$P_{ideal} = \frac{T^{3/2}}{2\pi\rho R^2}, \quad P_0 \approx \frac{1}{8} \rho c N_b C_{d0} \Omega^3 R^4, \quad (27)$$

with a typical value of γ in the order of 1.15. Eqs. (26) and (27) give a general overall design or layout basis of the rotor including all main quantities, such as weight (through the thrust), blade length, average chord, rotational speed and number of blades. An optimization procedure would be to make a design that minimizes the required power to generate a given thrust. However, note that this constitutes a first order approximate layout and that some of the parameters are constrained by other factors, such as aero-elastic properties, materials and noise restrictions.

In some cases, it is convenient to put the equations into dimensionless form. Dividing by the denominator of eq. (6), we get

$$C_{P0} = \frac{1/8\rho c N_b C_{d0} \Omega^3 R^4}{\frac{1}{2}\rho A (\Omega R)^3} = \frac{1}{4} \left(\frac{N_b c R}{A} \right) C_{d0} = \frac{1}{4} \left(\frac{N_b c}{\pi R} \right) C_{d0} = \frac{1}{4} \sigma C_{d0}, \quad (28)$$

and

$$C_P = \gamma C_{P,ideal} + C_{P0} = \frac{1}{2} \gamma C_T^{3/2} + \frac{1}{4} \sigma C_{D0}, \quad (29)$$

where $\sigma = \frac{N_b c}{\pi R}$ is the solidity.

1.2.3 Optimum plan form and twist distribution

Assuming small flow angles, the expression for thrust, eq. (20), may be approximated as

$$dT = N_b dF_z \simeq N_b f_l dr. \quad (30)$$

Introducing the lift coefficient, $C_l = \frac{f_L}{\frac{1}{2}\rho c (\Omega r)^2}$, in dimensionless form eq. (30) reads,

$$dC_T \simeq \frac{N_b \frac{1}{2}\rho c (\Omega r)^2 C_l}{\frac{1}{2}\rho \pi R^2 (\Omega R)^2} dr = \left(\frac{N_b c}{\pi R} \right) C_l \left(\frac{r}{R} \right)^2 d\left(\frac{r}{R} \right) = \sigma C_l y^2 dy, \quad (31)$$

where $y = \frac{r}{R}$ is the dimensionless radial distance along a rotor blade.

Employing axial momentum theory on an annulus of radial extent, dr , (see Fig. 2) a supplementary expression may be derived for the local thrust,

$$dT = d\dot{m} \cdot \Delta u = \rho v_i dA \cdot 2v_i = 4\pi \rho v_i^2 r dr. \quad (32)$$

In dimensionless form, the thrust reads,

$$dC_T = \frac{4\pi \rho v_i^2 r}{\frac{1}{2}\rho \pi R^2 (\Omega R)^2} dr = 8\lambda_i^2 y dy, \quad (33)$$

where $\lambda_i = \frac{v_i}{\Omega R}$ is the dimensionless induced velocity.

Combining eq. (31) and eq. (33), gives the following expression for the dimensionless induced velocity,

$$\lambda_i = \sqrt{\frac{\sigma y C_l}{8}}. \quad (34)$$

From ‘classical’ rotor theory, it is known that an optimum rotor operates at a constant induced velocity. This may be obtained by keeping a constant local angle of attack, resulting in a constant C_l -value along the rotor blade, and at the same time setting $\sigma y = \text{constant}$. Hence, from the definition of the solidity, we get the following expression for the optimum chord distribution,

$$c(r) = \frac{c_{tip}}{r}, \quad (35)$$

where c_{tip} denotes the chord length at the blade tip.

From Fig. (2) the angle of attack is found to be the difference between the local twist and flow angle, $\alpha = \theta - \phi$. Having a constant design angle of attack, α_D , we get $\theta = \alpha_D + \phi$.

Further, assuming small flow angles, we get

$$\phi = \tan^{-1} \left(\frac{v_i}{\Omega r} \right) \approx \frac{v_i}{\Omega r} = \frac{\lambda_i}{y}, \quad (36)$$

From eq. (15), the following relationship is obtained

$$\lambda_i = \frac{v_i}{\Omega R} = \sqrt{\frac{T}{2\rho A(\Omega R)^2}} = \frac{1}{2} \sqrt{C_T}. \quad (37)$$

Combining the above equations, the blade twist can be determined as

$$\theta(y) = \alpha_D + \frac{1}{2y} \sqrt{C_T}. \quad (38)$$

In the above analysis the optimum chord distribution, given by eq. (35), is defined except for the tip chord, which needs to be defined by other parameters. Likewise, the optimum twist, given by eq. (38), is defined except for the design angle of attack, which typically is taken as the value at which the drag-lift ratio, C_d / C_l , attains its minimum value. However, both quantities need to be defined by the designer. Another feature is that they both tend towards infinity when approaching the rotor axis. Therefore, the expressions are typically only used for the outer 70 % of the rotor blade. An alternative is to replace the hyperbolic distributions by linear expressions, e.g. by taking the slope at the tip and extrapolate that linearly towards the rotor center. For simplicity, the chord may also be taken as constant, leaving only the blade twist as a variable. These two alternatives will be exploited in the following.

1.2.4 Constant chord, untwisted blade design

Assuming now that a rotor is designed with a constant chord and a constant pitch angle θ_0 . Letting the rotor at design conditions operate in the linear range of the lift curve, the lift coefficient is given as

$$C_l \simeq \frac{dC_l}{d\alpha} \Big|_{\alpha=\alpha_0} (\alpha - \alpha_0) = C_{l,\alpha} (\alpha - \alpha_0), \quad (39)$$

where the slope of the lift curve, denoted as $C_{l,\alpha}$, is measured at the zero lift angle of attack.

Introducing eq. (39) into eq. (31), exploiting that $\alpha = \theta_0 - \phi$ and $\phi \simeq \frac{\lambda_i}{y}$, and integrating eq. (31), one obtains

$$C_T = \sigma C_{l,\alpha} \int_0^1 [(\theta_0 - \alpha_0) y^2 - \lambda_i y] dy = \sigma C_{l,\alpha} \left[\frac{(\theta_0 - \alpha_0)}{3} - \frac{\lambda_i}{2} \right]. \quad (40)$$

Since $\lambda_i = \frac{1}{2} \sqrt{C_T}$, we obtain the following expression for the pitch (twist) angle,

$$\theta_0 = 3 \left[\frac{C_T}{\sigma C_{l,\alpha}} + \frac{1}{4} \sqrt{C_T} \right] + \alpha_0. \quad (41)$$

1.2.5 Constant chord, linearly twisted blade design

Assuming now that a rotor is designed with a constant chord and a linear twist, given as $\theta(y) = a + by$, the thrust coefficient is determined as

$$C_T = \sigma C_{l,\alpha} \int_0^1 [(a + by - \alpha_0) y^2 - \lambda_i y] dy = \sigma C_{l,\alpha} \left[\frac{(a - \alpha_0)}{3} + \frac{b}{4} - \frac{\lambda_i}{2} \right]. \quad (42)$$

We are here facing the problem that there are two unknowns, a and b , in the expression. A way to circumvent the problem is to assume a hyperbolic optimum twist distribution taking the form $\theta(y) = \frac{\theta_{tip}}{y}$, and replace it by a linear expression by taking the slope at the tip and extrapolate it linearly towards the rotor center. Evaluating the twist and twist slope for $y = 1$, shows that $a = 2\theta_{tip}$ and $b = -\theta_{tip}$, resulting in the simple linear expression,

$$\theta = \theta_{tip} (2 - y). \quad (43)$$

Inserting the constants a and b in eq. (42), exploiting that $\lambda_i = \frac{1}{2} \sqrt{C_T}$, the twist at the tip is determined as,

$$\theta_{tip} = \frac{12}{5} \left[\frac{C_T}{\sigma C_{l,\alpha}} + \frac{1}{4} \sqrt{C_T} + \frac{1}{3} \alpha_0 \right]. \quad (44)$$

1.2.6 Constant chord, optimum twist

Although helicopters mainly are designed for hover, they spend a large part of their time in forward flight. Hence, a fully optimal design covering all flight duties is not possible and therefore the design in most cases is limited to constant chord and linear twist. However, as a design reference an optimum twist distribution, as the one exploited in section 2.1.5, takes the

form, $\theta(y) = \frac{\theta_{tip}}{y}$. From eqs. (31) and (39), we get

$$\begin{aligned} C_T &= \sigma C_{l,\alpha} \int_0^1 [\theta - \phi - \alpha_0] y^2 dy = \sigma C_{l,\alpha} \int_0^1 [(\theta_{tip} - \lambda_i) y - \alpha_0 y^2] dy \\ &= \sigma C_{l,\alpha} \left[\frac{(\theta_{tip} - \lambda_i)}{2} - \frac{\alpha_0}{3} \right] \end{aligned} \quad , \quad (45)$$

from which the tip twist is determined as

$$\theta_{tip} = 2 \frac{C_T}{\sigma C_{l,\alpha}} + \frac{1}{2} \sqrt{C_T} + \frac{2}{3} \alpha_0. \quad (46)$$

1.3 Axial momentum theory in axial climb and descent

Assuming that the rotor now is climbing with a velocity V_c , the relative velocity far upstream will point downstream and all induced velocities will be superposed by V_c (see Fig. 3) .

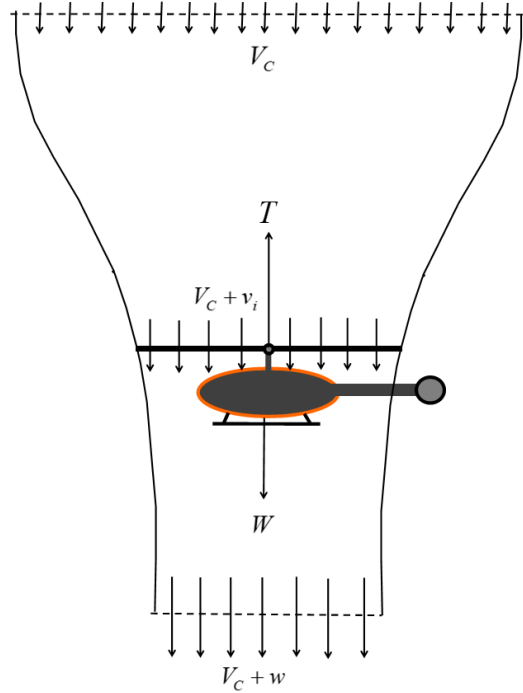


Fig. 3. Control volume and flow characteristics of actuator disc model of climbing rotor.

Denoting by $\dot{m}$ the mass flow through the rotor plane, as for the hovering case, mass conservation (eq. 1) gives the following integral relations,

$$\dot{m} = \int_{disc} \rho \mathbf{V} \cdot d\mathbf{A} = \int_{\infty} \rho \mathbf{V} \cdot d\mathbf{A} . \quad (47)$$

At the plane of the rotor, Assuming one-dimensional flow theory and denoting the by w the outflow velocity in the wake, we get the following relationship,

$$\dot{m} = \rho A (V_C + v_i) = \rho A_{\infty} (V_C + w), \quad (48)$$

where A_{∞} denotes the area of the outflow plane.

Applying axial momentum balance over the actuator disc, eq. (2), we get

$$T = \int_{\infty} u \rho \mathbf{V} \cdot d\mathbf{A} - \int_0 u \rho \mathbf{V} \cdot d\mathbf{A}, \quad (49)$$

where the contribution of the pressure disappears as it can be shown that its net contribution on the control volume is zero. Since the climb velocity corresponds to the relative incoming velocity, both integrals are finite, and we get

$$T = \dot{m} (V_C + w) - \dot{m} V_C = \dot{m} w. \quad (50)$$

From energy conservation, eq. (4), the consumed power reads

$$\begin{aligned} T(V_C + v_i) = P &= \int_{\infty} \frac{1}{2} u^2 \rho \mathbf{V} \cdot d\mathbf{A} - \int_0 \frac{1}{2} u^2 \rho \mathbf{V} \cdot d\mathbf{A} \\ &= \frac{1}{2} \dot{m} (V_C + w)^2 - \frac{1}{2} \dot{m} V_C^2 = \frac{1}{2} \dot{m} w (2V_C + w) \end{aligned} \quad (51)$$

Combining eqs. (50) and (51), we get

$$P = \dot{m} w (V_C + v_i) = \frac{1}{2} \dot{m} w (2V_C + w), \quad (52)$$

resulting in the relationship,

$$v_i = \frac{1}{2} w, \quad (53)$$

showing that the induced velocity in the rotor plane is equal to half of the induced velocity in the far wake, as in the case of the hovering rotor. Exploiting this in eq. (51), gives the result,

$$T = \dot{m} w = 2\rho A (V_C + v_i) v_i. \quad (54)$$

Letting the induced velocity in hover represent a reference velocity,

$$v_H \equiv \sqrt{\frac{T}{2\rho A}}. \quad (55)$$

from eq. (54) we obtain

$$\left(\frac{v_i}{v_H}\right)^2 + \left(\frac{V_C}{v_H}\right)\left(\frac{v_i}{v_H}\right) - 1 = 0. \quad (56)$$

Solving this for the dimensionless induced velocity gives two possible solutions,

$$\frac{v_i}{v_H} = -\left(\frac{V_C}{2v_H}\right) \pm \sqrt{\left(\frac{V_C}{2v_H}\right)^2 + 1}, \quad (57)$$

where the positive sign in front of the square root is chosen, as it is the physical one given a positive value of the induced velocity.

$$\frac{v_i}{v_H} = -\left(\frac{V_C}{2v_H}\right) + \sqrt{\left(\frac{V_C}{2v_H}\right)^2 + 1}, \quad (58)$$

From the equation it is seen that the induced velocity decreases when the climb velocity increases. An alternative form, is to divide eq. (54) with the denominator of eq. (5) and introducing the thrust coefficient,

$$C_T = \frac{T}{\frac{1}{2}\rho A(\Omega R)^2} = 4\left(\frac{v_i}{V_C}\right)\left[1 + \left(\frac{v_i}{V_C}\right)\right]\left(\frac{V_C}{\Omega R}\right)^2. \quad (59)$$

Introducing the advance ratio, $\lambda_c = \frac{V_C}{\Omega R}$, eq. (60) is written as,

$$\left(\frac{v_i}{V_C}\right)^2 + \left(\frac{v_i}{V_C}\right) - \frac{C_T}{4\lambda_c^2} = 0, \quad (60)$$

from which the induced velocity can be determined,

$$\frac{v_i}{V_C} = \frac{1}{2}\left(-1 \pm \sqrt{1 + \frac{C_T}{\lambda_c^2}}\right), \quad (61)$$

where the positive sign is chosen, as v_i always has to be positive in climb velocity,

$$\frac{v_i}{V_C} = \frac{1}{2} \left(-1 + \sqrt{1 + \frac{C_T}{\lambda_C^2}} \right). \quad (62)$$

As long as the climb velocity is positive, the rotor operates in the so-called normal propeller state. However, when the rotor goes into a descent state, i.e. when V_C becomes negative, it is not clear if the slipstream is positive or negative. This uncertainty actually violates the uniqueness of the model. In the range $-1 \leq \frac{V_C}{2v_H} \leq 0$ the velocity in the slipstream may be

both positive and negative, and axial momentum theory ceases to work. This range is dominated by turbulence and the generation of large scale recirculating vortices, and should be avoided as it causes an unstable flight condition. For even lower values of the descent velocity, i.e. for $\frac{V_C}{2v_H} < -1$, the rotor goes into the so-called windmill state, where the rotor starts to act as a wind turbine, meaning that it extracts energy from the flow. In this state the induced velocity changes direction, as compared to the thrust, and the axial momentum equation reads,

$$T = -\dot{m}w = -2\rho A(V_C + v_i)v_i, \quad (63)$$

leading to the equation

$$\frac{T}{2\rho A} = v_H^2 = -V_C v_i - v_i^2 \Rightarrow \left(\frac{v_i}{v_H} \right)^2 + \left(\frac{V_C}{v_H} \right) \left(\frac{v_i}{v_H} \right) + 1 = 0, \quad (64)$$

with the solution

$$\frac{v_i}{v_H} = -\left(\frac{V_C}{2v_H} \right) - \sqrt{\left(\frac{V_C}{2v_H} \right)^2 - 1} \quad \text{for} \quad \frac{V_C}{2v_H} < -1. \quad (65)$$

The negative sign in front of the square root has been used in order ensure that the induced velocity is positive and goes to zero when $V_C \rightarrow -\infty$.

As in the propeller state, the solution may be written in terms of the thrust coefficient,

$$C_T = \frac{T}{\frac{1}{2}\rho A(\Omega R)^2} = -4 \left(\frac{v_i}{V_C} \right) \left[1 + \left(\frac{v_i}{V_C} \right) \right] \left(\frac{V_C}{\Omega R} \right)^2, \quad (66)$$

which can be written as,

$$\left(\frac{v_i}{V_c}\right)^2 + \left(\frac{v_i}{V_c}\right) + \frac{C_T}{4\lambda_c^2} = 0. \quad (67)$$

Again two solutions exists, and here the one with a plus in front of the square root is chosen,

$$\frac{v_i}{V_c} = \frac{1}{2} \left(-1 + \sqrt{1 - \frac{C_T}{\lambda_c^2}} \right). \quad (68)$$

To illustrate the various flow states, Fig. 4 shows the induced velocity as function of climb velocity. The normal propeller state, eq. (58), is here shown to the right for $\frac{V_c}{v_H} > 0$, and the

windmill brake state, eq. (65), to the left for $\frac{V_c}{v_H} < -2$. In between one observes the vortex

ring state where momentum theory is no longer valid. Different measurements have shown this state to be unstable, as illustrated by the erratic nature of the inferred measurements. This range furthermore constitutes a safety issue as it can lead to loss of rotor control.

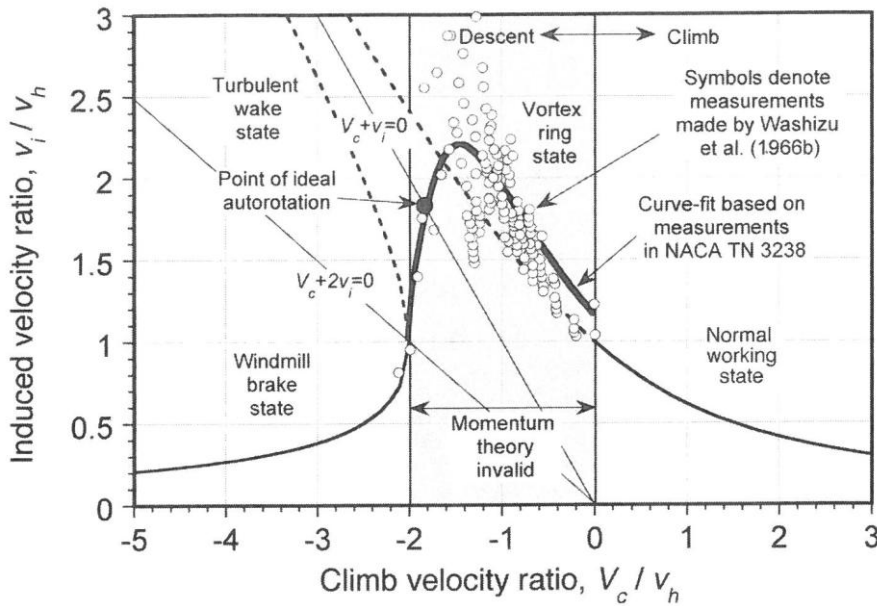


Fig. 4. Illustration of induced velocity as function of climb and descent velocity (from Leishman (2000)).

1.4 Momentum theory for helicopter in forward flight

In forward flight conditions, the rotor needs to be tilted forward in order to produce a forward pointing force. Furthermore, to maintain a constant and steady flight condition the horizontal component of the thrust must be in balance with the drag of the helicopter and the vertical component of the thrust must outbalance the weight. Defining by β the tilt angle between the rotor plane and the horizontal, we get

$$T \sin \beta = D, \quad \text{and} \quad T \cos \beta = W, \quad (69)$$

from which we get

$$\beta = \tan^{-1}\left(\frac{D}{W}\right) \quad \text{and} \quad T = \sqrt{W^2 + D^2} . \quad (70)$$

Fig. 5 shows a flow model for momentum analysis of a rotor in horizontal forward flight.

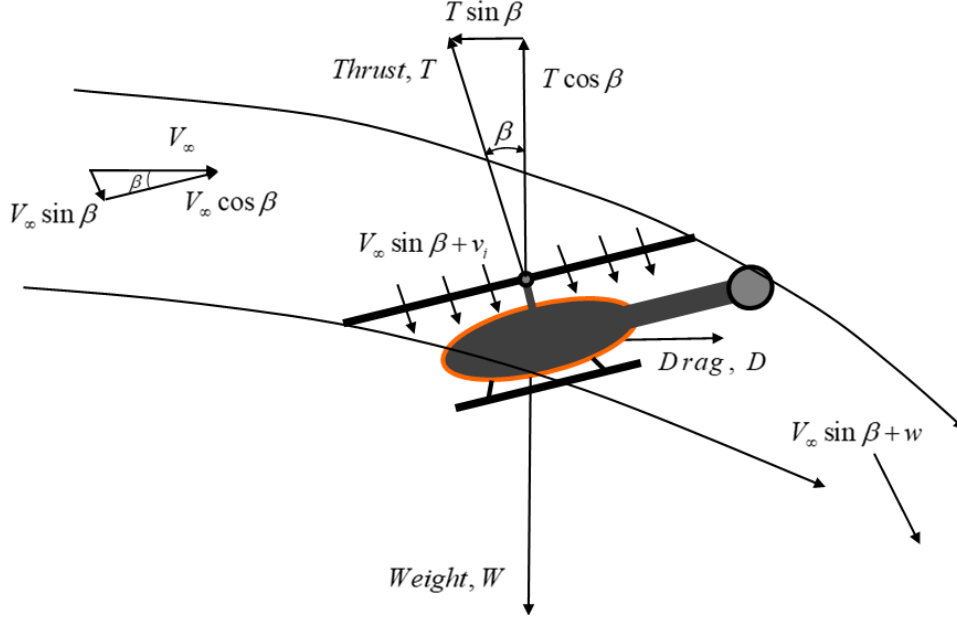


Fig. 5. Control volume and geometrical/kinematic properties for rotor in horizontal forward flight.

To compute the power using momentum theory requires some approximate assumptions that cannot be fully justified. Assuming that the rotor moves forward in the horizontal direction with velocity V_∞ . Hence, far upstream of the rotor, the resulting velocity can be decomposed in a velocity perpendicular to the rotor plane, $V_\infty \sin \beta$, and a component parallel to the rotor plane, $V_\infty \cos \beta$. Due to the action of the thrust, an induced velocity, v_i , appears in the rotor plane, and similarly a velocity, w , is induced in the wake, as seen in Fig. 5. At the rotor plane, the resultant velocity is given as

$$U = \sqrt{(V_\infty \cos \beta)^2 + (V_\infty \sin \beta + v_i)^2} . \quad (71)$$

A major assumption, as was already formulated in the pioneering work of Glauert (1935), is that the mass flow through the rotor disk takes the same expression as in axial flight, meaning that

$$\dot{m} = \rho A U = \rho A \sqrt{(V_\infty \cos \beta)^2 + (V_\infty \sin \beta + v_i)^2} . \quad (72)$$

This equation can actually not be proven rigorously, but combined with momentum theory (see eq. 76) it reverts to the correct form for a helicopter in climb, and at the other extreme, when the horizontal velocity is large compared with the induced velocity, it gives the result

corresponding to an inclined wing of semi-span R with constant downwash. To determine the induced velocity, the same technique as used for hover and axial climb is utilized. Applying axial momentum theory in the direction normal to the rotor disk, the thrust is

$$T = \dot{m}(V_\infty \sin \beta + w) - \dot{m}V_\infty \sin \beta = \dot{m}w. \quad (73)$$

Multiplying this by the velocity through the rotor disc, we get the power,

$$\begin{aligned} P &= T(V_\infty \sin \beta + v_i) = \frac{1}{2}\dot{m}(V_\infty \sin \beta + w)^2 - \frac{1}{2}\dot{m}V_\infty \sin^2 \beta \\ &= \frac{1}{2}\dot{m}w(2V_\infty \sin \beta + w) \end{aligned} \quad (74)$$

Combining eq. (73) and (74), gives the expected result, that the induced velocity in the rotor plane, as usual, is half of wake induction,

$$V_\infty \sin \beta + v_i = \frac{1}{2}(2V_\infty \sin \beta + w) \Rightarrow v_i = \frac{1}{2}w. \quad (75)$$

From momentum analysis the thrust is found as

$$T = 2\rho AUv_i = 2\rho Av_i \sqrt{(V_\infty \cos \beta)^2 + (V_\infty \sin \beta + v_i)^2}. \quad (76)$$

With reference to the hover case, an induced reference velocity is given as

$$v_H = \sqrt{\frac{T}{2\rho A}}, \quad (77)$$

and, by combining the equations, the actual induced velocity is determined as,

$$v_i = \frac{v_H^2}{\sqrt{(V_\infty \cos \beta)^2 + (V_\infty \sin \beta + v_i)^2}}. \quad (78)$$

This equation may be solved iteratively using e.g. a Newton-Raphson procedure. Alternatively, it may be rewritten into the following polynomial form,

$$v_i^4 + 2V_\infty \sin \beta \cdot v_i^3 + V_\infty^2 \cdot v_i^2 - v_H^4 = 0, \quad (79)$$

which can be solved directly to give the desired induced velocity. Unfortunately, this gives four solutions, of which only one makes sense. The subsequent power is determined as

$$P_{ideal} = T \cdot (V_\infty \sin \beta + v_i), \quad (80)$$

which is without losses. With losses we get

$$P = \kappa P_{ideal} + P_o = \kappa P_{ideal} + \frac{1}{8} \rho N_b \Omega^3 C_{do} \left(\frac{c}{R} \right) R^5. \quad (81)$$

In many cases it is convenient to put the equations in dimensionless form by introducing the following dimensionless numbers:

$$\text{Induced inflow ratio: } \lambda_i = \frac{v_i}{\Omega R}$$

$$\text{Inflow ratio:} \quad \lambda_{\text{inf}} = \frac{V_{\infty} \sin \beta + v_i}{\Omega R}. \quad (82)$$

$$\text{Advance ratio:} \quad \lambda_{\text{adv}} = \frac{V_{\infty}}{\Omega R}$$

Inserting these into eq. (78), recalling that $\lambda_H \equiv \frac{v_H}{\Omega R} = \frac{1}{2} \sqrt{C_T}$, we get two alternative forms to eq. (78),

$$\lambda_i = \frac{C_T}{4\sqrt{\lambda_{\text{adv}}^2 + \lambda_i^2 + 2\lambda_{\text{adv}} \sin \beta \cdot \lambda_i}}. \quad (83)$$

$$\lambda_{\text{inf}} = \lambda_{\text{adv}} \sin \beta + \frac{C_T}{4\sqrt{\lambda_{\text{inf}}^2 + \lambda_{\text{adv}}^2 \cos^2 \beta}}. \quad (84)$$

As for eq. (78), these equations can be solved iteratively for the unknown inflow ratio or directly as a 4th order polynomial. Input to the equations is in all cases, the thrust coefficient, corresponding to the weight, the advance ratio and the inclination of the rotor.

The above equations are valid for horizontal flight. However, the equations can easily be extended to a more general form in the case that forward flight is combined with climbing. Assuming the helicopter being climbing with vertical velocity, U_{∞} , the total dimensionless induced velocity in the rotor plane now reads,

$$\lambda_{\text{inf}} = \frac{V_{\infty} \sin \beta + v_i + U_{\infty} \cos \beta}{\Omega R}. \quad (85)$$

Assuming furthermore that the mass flow through the rotor is invariant, eq. (84) reads,

$$\lambda_{\text{inf}} = \lambda_{\text{adv}} \sin \beta + \lambda_C \cos \beta + \frac{C_T}{4\sqrt{\lambda_{\text{inf}}^2 + \lambda_{\text{adv}}^2 \cos^2 \beta}}, \quad (86)$$

where $\lambda_C = \frac{U_{\infty}}{\Omega R}$ is the advance ratio perpendicular to the rotor plane.

1.5 Blade-element momentum theory

As the name indicates, the blade-element momentum (BEM) theory is a combination of axial momentum theory and blade-element analysis. Consider an element of the blade of radial extent Δr (see Fig. 2), from momentum theory the local thrust contribution is given as

$$\Delta T_{\text{mom}} = \Delta \dot{m} \cdot (V_w - V_C) = \rho (V_C + v_i) 2\pi r \Delta r \cdot 2v_i = 4\pi \rho (V_C + v_i) v_i r \Delta r. \quad (87)$$

From blade-element analysis (cf. eq. 20), we get

$$\begin{aligned}\Delta T_{BE} &= N_b dF_z = N_b (f_L \cos \phi - f_D \sin \phi) \Delta r \\ &= \frac{1}{2} N_b \rho c V_{rel}^2 [C_l \cos \phi - C_d \sin \phi] \Delta r\end{aligned}\quad (88)$$

where the flow angle $\phi = \tan^{-1} \left(\frac{u_i}{\Omega r} \right)$ and the local relative velocity

$V_{rel} = \sqrt{(V_C + u_i)^2 + (\Omega r)^2}$. Neglecting the induced velocity component in the tangential direction, the only unknown is the induced axial velocity, $u_i(r)$. To determine this, the contribution from momentum theory needs to be modified with the tip correction, which essentially accounts for the impact of a finite number of blades in the momentum theory, $\Delta T_{BE} = F \cdot \Delta T_{mom}$, where we here use Glauert's version of the Prandtl tip correction (Glauert, 1935):

$$F = \frac{2}{\pi} \cos^{-1} [\exp(-f)], \quad \text{where} \quad f = \frac{N_b}{2} \left[\frac{R-r}{r \sin \phi} \right]. \quad (89)$$

Although there is only one unknown in the above system of equations, it is nonlinear and includes look-up in tables to determine the airfoil data, and therefore an iterative solution technique is required. There are different techniques for solving the system, ranging from simple updating to Newton's method. Here we choose a simple iterative technique similar to bi-section, which is to update the unknown induced velocity from the following scheme:

$$v_i^{n+1} = v_i^n + C \cdot [\Delta T_{BE} - F \cdot \Delta T_{mom}], \quad (90)$$

where $()^n$ is an iteration count, and C is a constant which here is set equal to $\frac{1}{2}$. As starting guess for the induced velocity, one may put it equal to zero or use the value from the one-dimensional momentum analysis. For convergence, we employ the criterion that convergence is obtained when $\|\Delta T_{BE} - F \cdot \Delta T_{mom}\| < 10^{-6}$. The solution procedure is carried out for each blade section, and the total thrust is obtained by summing up all the components. Likewise, the power demand is determined by adding up the power contribution from each blade element,

$$\Delta P = \frac{1}{2} N_b \Omega \rho c V_{rel}^2 [C_l \sin \phi + C_d \cos \phi] r \Delta r. \quad (91)$$

In many case it is convenient to reformulate the equations using dimensionless quantities. Introducing a local thrust coefficient by dividing the thrust, eq. (87), by the denominator of eq. 5, we get

$$\Delta C_{T,mom} = \frac{\Delta T_{mom}}{\frac{1}{2} \rho \pi R^2 (\Omega R)^2} = 8 (\lambda_c + \lambda_i) \lambda_i y \Delta y. \quad (92)$$

Likewise, eq. (88) is reformulated as

$$\Delta C_{T,BE} = \frac{\Delta T_{BE}}{\frac{1}{2} \rho \pi R^2 (\Omega R)^2} = \sigma \left((\lambda_c + \lambda_i)^2 + y^2 \right) [C_l \cos \phi - C_d \sin \phi] \Delta y, \quad (93)$$

where $\sigma = \frac{N_b c}{\pi R}$ is the solidity and $y = \frac{r}{R}$ is the dimensionless radial coordinate. Using the same procedure as in eq. (90), an iterative solution procedure takes the form,

$$\lambda_i^{n+1} = \lambda_i^n + C \cdot [\Delta C_{T,BE} - F \cdot \Delta C_{T,mom}], \quad (94)$$

where, as in eq. (90), $()^n$ is the iteration count, C is a constant, and F is the tip correction with $f = \frac{N_b}{2} \left[\frac{1-y}{y \sin \phi} \right]$.

One of the main complications in the above procedure is the use of tabulated airfoil data as they are nonlinear when operating near and after stall. As an example, consider the polar for the S826 airfoil (see Fig. 6), which exhibits a linear behavior around the zero-lift angle ($\alpha \approx -5^\circ$), but becomes strongly nonlinear around stall ($\alpha \approx 10^\circ$).

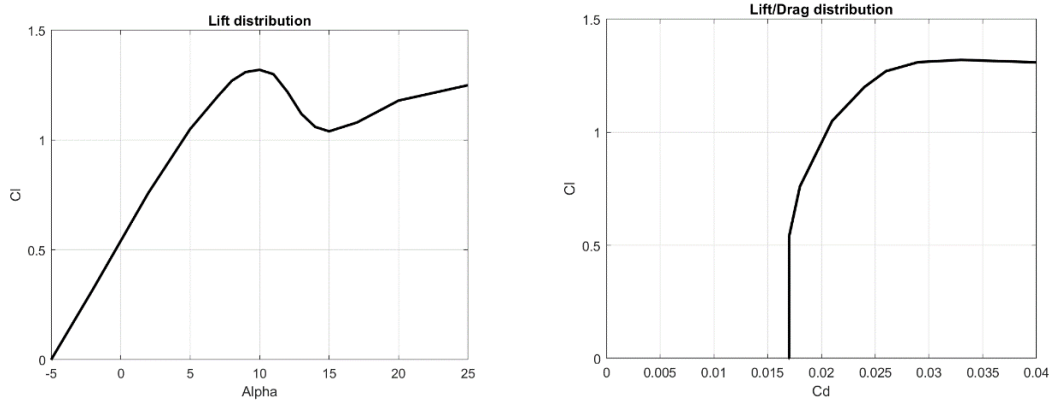


Fig. 6. Distributions of lift coefficients and lift/drag ratio for the S826 airfoil at a Reynolds number of 300,000.

When operating in the linear regime and at small angles of attack, the equations forming the BEM model can be linearized and a simplified solution algorithm can be devised.

1.6 Linearized BEM model

In most part of the operational envelope, the rotor operates in the linear regime of the lift coefficient and at small flow angles. In this case, the following approximations are valid

$$\cos \phi \approx 1, \quad \sin \phi \approx \phi, \quad \tan \phi \approx \phi, \quad C_d \ll C_l.$$

Exploiting these approximations, and combining eqs. (92) and (93), we get the following

$$\Delta C_{T,BE} = F \Delta C_{T,mom} \Rightarrow \sigma y^2 C_l \approx 8F (\lambda_c + \lambda_i) \lambda_i y. \quad (94)$$

In the linear regime, the lift coefficient can be written as

$$C_l = C_{l,\alpha} (\alpha - \alpha_0) = C_{l,\alpha} (\theta - \phi - \alpha_0). \quad (95)$$

Utilizing that $\phi = \frac{\lambda_i}{y}$, combining eqs. (94) and (95), we get

$$\lambda_i^2 + \left(\lambda_c + \frac{\sigma C_{l,\alpha}}{8F} \right) \lambda_i - \frac{\sigma C_{l,\alpha} (\theta - \alpha_0) y}{8F} = 0. \quad (96)$$

Solving this for the unknown induced velocity, we get

$$\lambda_i = -\frac{1}{2} \left[\lambda_c + \frac{\sigma C_{l,\alpha}}{8F} - \sqrt{\left(\lambda_c + \frac{\sigma C_{l,\alpha}}{8F} \right)^2 + \frac{\sigma C_{l,\alpha} (\theta - \alpha_0) y}{2F}} \right]. \quad (97)$$

The above equation includes climb through the vertical advance ratio. Considering hover conditions, where $\lambda_c = 0$, eq. (97) can be reduced further to the simple equation,

$$\lambda_i = \frac{\sigma C_{l,\alpha}}{16F} \left[\sqrt{1 + \frac{32F}{\sigma C_{l,\alpha}} (\theta - \alpha_0) y} - 1 \right]. \quad (98)$$

As the tip correction, F , contains the flow angle, and by this the induced velocity, an iterative solution procedure is required. As a first guess the tip correction may be ignored by setting $F=I$. Next, the equation is updated by including the tip correction, and the iteration continues until convergence is achieved.

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